

Customer Churn Prediction & LTV Engine

Week 4 • Day 3

Advanced Dashboard — Customer Risk Table & Filters

Build interactive drill-downs, filters, and the at-risk customer list

■ Day Objectives

- ◆ Create a filterable customer risk score table
- ◆ Add dashboard filters (LTV segment, contract type, risk level)
- ◆ Build the 'Revenue at Risk' chart
- ◆ Create an executive summary dashboard tab
- ◆ Export dashboard as PDF for stakeholders

■ 1. Customer Risk Table

■ High-Risk Customer List SQL — paste in Metabase Native Query

```
SELECT
  customerid,
  contract,
  internetservice,
  tenure,
  ROUND(monthlycharges::numeric, 2) AS monthly_charges,
  ROUND(churn_probability::numeric, 3) AS churn_prob,
  CASE
    WHEN churn_probability > 0.7 THEN 'HIGH RISK'
    WHEN churn_probability > 0.4 THEN 'MEDIUM RISK'
```

```

        ELSE 'LOW RISK'
    END AS risk_level,
    ROUND(total_ltv::numeric, 2) AS total_ltv,
    ltv_segment
FROM customers_ltv
ORDER BY churn_probability DESC
LIMIT 100;

```

■ 2. Revenue at Risk Chart

■ Revenue at Risk by Segment SQL

```

SELECT
    ltv_segment,
    CASE WHEN churn_probability > 0.4 THEN 'At Risk' ELSE 'Stable' END AS status,
    COUNT(*) AS customers,
    ROUND(SUM(total_ltv)::numeric, 2) AS total_ltv_at_risk
FROM customers_ltv
GROUP BY ltv_segment, status
ORDER BY ltv_segment, status;

```

■ 3. Commit & Checklist

- Customer risk table created with 100 rows sortable by churn probability
- Revenue at Risk chart shows \$ value by segment
- Dashboard has at least 3 filter controls

■ Week4-Day3 Commit

```

git add .
git commit -m "Week4-Day3: Advanced dashboard – risk table, revenue at risk, filters"
git push origin main

```

■ Estimated Time: 5 – 6 hours

— End of Week 4 Day 3 — Zaalima Development Confidential —